PIC 10A 1A

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Today…

- HW6 Hints
- Exercise Problems Related to HW6

HW6 Hints

HW6 Hints (descending.cpp)

- Recall that, we wrote the array sort function that sorts an array in descending order
  - Week7_Array_Functions_Sol.cpp

- You can do the exact same thing with vectors!

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HW6 Hints (mostfrequent.cpp)

- Many strategies out there
- If we want to preserve the order of the letters as in the HW instruction
  - i.e. input: “howdy” → output: “howdy” (not “dhowy”, a sorted string)
- Then it becomes a little more complex, but not too much
  - So let’s solve it in this way

- Suggestion: Count the frequency first, and construct a string later
  - Step 1. Count the frequencies of each letter
  - Step 2. Find the maximum frequency
  - Step 3. Construct a new string, with the letters with maximum frequency

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HW6 Hints (mostfrequent.cpp)

- Step 1. Count the frequencies of each letter
  - Say, you have `vector<int> freq(nPossibleLetters);`

- Step 2. Find the maximum frequency
  - `int max_freq = // max frequency in freq`

- Step 3. Construct a new string, with the letters with maximum frequency

```
string answer;
for ( each character c in input ) {
    if (freq of c == max_freq) { // c has max frequency
        answer += c; // append
        // no repetition of c should be made later
        // e.g. “rgm” instead of “rgrmm” for “programming”
        // what should we do?
    }
}
```

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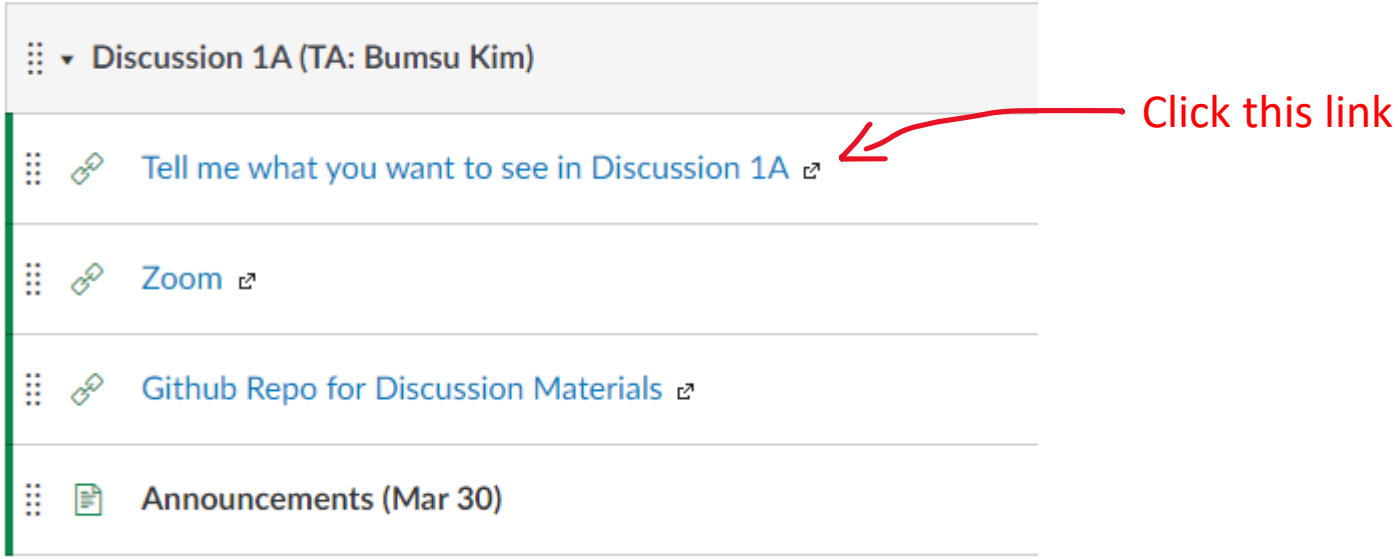
Exercise Problems for HW6

+ Review of Lecture Materials

[See Live Coding]

Your Feedback is welcome

- Don’t hesitate to give a feedback on the discussion
- You can use a link in BruinLearn (Google Form)